

Auditory VR Generative System for Non-Experts to Reproduce Human Memories Through Natural Language Interactions

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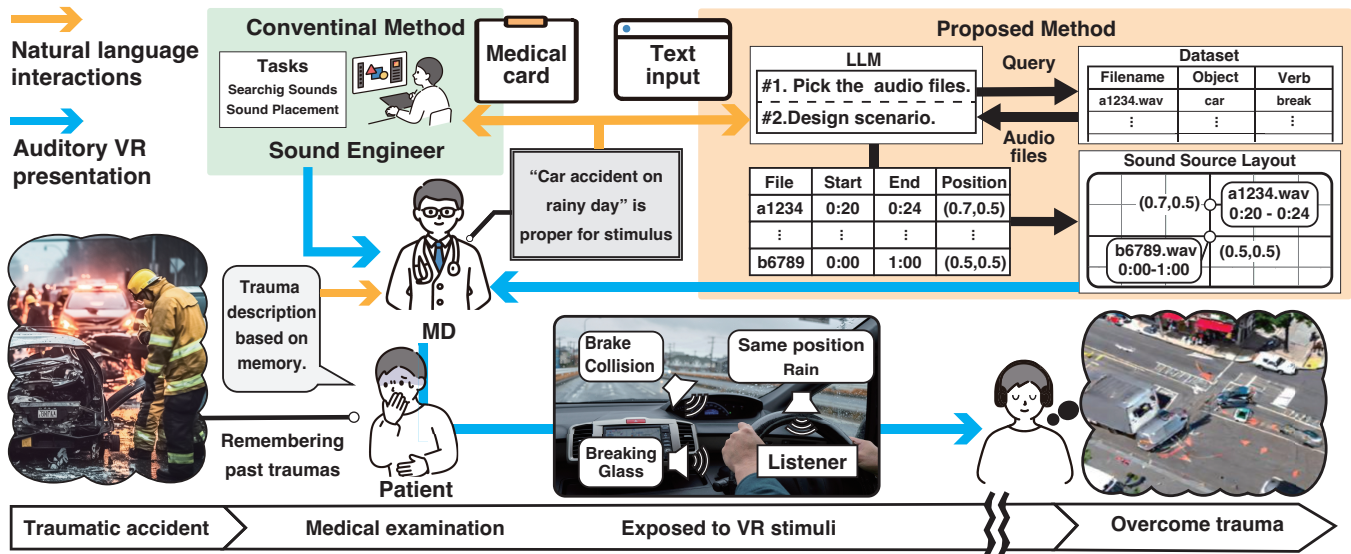


Figure 1: Treatment flow for Exposure Therapy: Patients report details of past traumatic events to the medical doctor. Conventionally, the doctor would create a chart for an engineer, who would then manually produce an auditory VR. However, in our proposed method, the doctor directly produce an auditory VR through natural language interactions with the system.

ABSTRACT

We propose an automatic auditory VR generative system based on natural language input and attempt to apply it to VR exposure therapy, a promising treatment for Post-Traumatic Stress Disorder (PTSD). The system consisted of the user interface, developed based on the Large Language Model (LLM), the auditory event dataset, which has metadata of “subject” and “verb” and spatial audio generator.

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1 INTRODUCTION

Along with the development of VR content, demand for auditory VR, as typified by stereophonic sound, has been increasing. “Auditory VR” refers to technology that localizes sound to a specific position and gives the user the sensation of being in reality. However, creating advanced auditory VR requires a wide range of specialized knowledge and skills and takes considerable time. Existing studies on auditory VR propose a framework that generates spatial audio from video and monaural sound by machine learning[Zhou et al. 2020]. But, creating arbitrary situations requires the selection of video and audio to suit individual users thus, it cannot be said that this has reduced the effort required. So we propose a system that automatically generates auditory VR from natural language interaction and can be easily handled by non-experts.

One application of this system is the automatic generation of VR stimulation for VR exposure therapy, a promising treatment for PTSD. This disorder is high prevalence, and trauma-focused cognitive-behavioral therapy, which encourages exposure to traumatic memories, is the most effective treatment and is recommended in various guidelines[Lee et al. 2016]. This therapy involves an element of exposure but also causes emotional distress to the

patient, often leading to dropout from treatment. Therefore, VR exposure therapy is being studied to induce habituation (decrease in anxiety through repeating) by reproducing the patient's traumatic events in VR and repeatedly exposing them to them. This therapy has the advantage of allowing treatment at any place and time, without the need to go out to be exposed to an unsettling situation or to perform labor-intensive procedures such as recounting the traumatic event in detail. In addition, this therapy is non-inferior to existing PTSD treatments [Kothgassner et al. 2019]. However, despite its high therapeutic efficacy and safety, VR exposure therapy has not become widespread. One of the reasons for the limited adoption of VR exposure therapy is the need for engineers to manually create VR stimuli according to the trauma of each patient. Our proposed system simplifies this process by leveraging natural language processing, enabling the quick generation of auditory VR without the need for specialized expertise, thereby addressing the issues above.

2 OUR APPROACH

As shown in Fig. 1, our proposed system allows medical doctors to create auditory VR for therapy directly from natural language interactions. Central to this system is the LLM (GPT-4), which we associated with the dataset using llama-index and langchain. The system selects several suitable audio files based on the medical doctor's input from the dataset. It then refines this selection, suggesting scenarios and sound source positions. As shown in Fig. 2, users can provide feedback in natural language to improve these suggestions. Finally, the system applies a head-related transfer function to each audio file using the localization position, then mixes the sounds based on playback time to produce the auditory VR output.

This paper aims to reproduce the situation of a "car accident on a rainy day" using natural language input. We created our own dataset, as shown in Fig. 1, and assigned two labels to each audio file: "subject" (what the sound source is) and "verb" (what the source is doing) (ex. subject: car, verb: braking). We intend this labeling to enhance the accuracy of queries, and the use of a label that integrates "subject" and "verb" (ex. car_braking) does not produce the expected results. Although a combination of subject sounds and scene sounds is known to enhance the sense of presence [Mashiba et al. 2022], labeling scene sounds in the form of "subject" and "verb" is challenging. Therefore, the "subject" was defined as "ambient," and its characteristics were described by "verb" (ex. subject: ambient, verb: rain). By using separated labels and changing internal prompts slightly, it becomes possible to pick the necessary sounds. We ran an experiment where the system explored a dataset that contained about one-third of relevant data. As a result, only about 35% of the relevant sounds were captured using the integrated labels, and occasionally the most important sounds, rain and collision, were not captured. On the other hand, using separated labels enabled more than 50% of the sounds to be captured, ensuring the capture of rain and crash sounds. Although irrelevant sounds are included at this stage, unnecessary sounds are eliminated at the scenario and position determination stages. So by using this method, the auditory VR output becomes high quality. To evaluate the system, we operated the system with natural language input and recorded how the user received auditory VR after several exchanges. For the

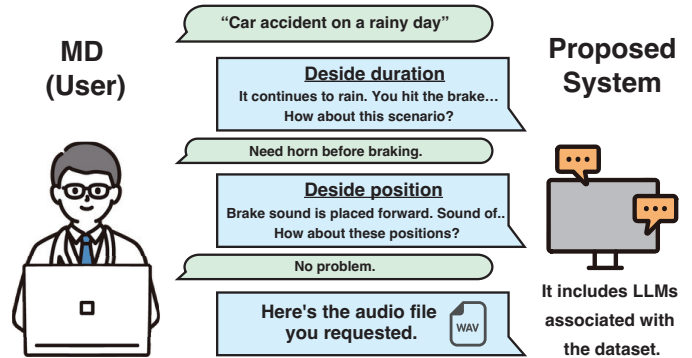


Figure 2: An example of the system and user interaction

evaluation of our system's time performance, we ran the system five times and measured the time taken from input to output. This was conducted on a 1.4 GHz quad-core Intel Core i5 CPU with 16 GB of memory and a communication speed of about 95 Mbps.

3 RESULT & DISCUSSION

The medical doctors reviewed that natural language input allows independent operation without the knowledge of technology or engineering, and they can get the final output. However, they also pointed out that while the quality of the output auditory VR met the standards for a prototype, realism needs improvement to be suitable for therapeutic applications. In our time measurements, the average duration to produce a auditory VR was 5 minutes and 25 seconds: about 2 minutes for user interaction, about 3 minutes for system response, and 15 seconds for speech synthesis. In contrast, while there was a difference in not having some of the source data on hand, according to an engineer with experience creating a similar auditory VR, it took about five hours to create. From this result, we can say this system allows us to create auditory VR much faster and easier than manual creation, and its quality holds potential for practical use in medical settings. However, VR exposure therapy uses multiple auditory VR, and creating all of these requires a reasonable amount of time even using the system. Therefore, in the future, we are considering using a patient's medical chart as input. From this single input, the system analyzes which situations are most effective when presented with auditory VR in which order and, based on this analysis, generate the necessary multiple stimuli.

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